BOAMP Layer 2 Technical Report

Enriched buyer identity sensitivity layer

Current independent study

End-to-end refresh: 2026-07-28

Every figure below was re-derived from the current artifacts; counts and link identities are unchanged from the 2026-07-27 run, which was verified by the frozen parity fingerprints in `tests/fixtures/baseline_fingerprints.json`.

Abstract

Layer 2, `enriched`, keeps the exact same 3,380 source contracts as Layer 1 but upgrades buyer identity with validated SIREN enrichment and a conservative historical alias bridge. It is a sensitivity layer, not a replacement for the BOAMP-native specification. The current run produces 13,524 candidate pairs over 2,165 sources and 1,188 balanced proxy-renewal events at the shared balanced threshold 0.343167. Relative to Layer 1, it adds 189 source links, removes 4, and changes 37 selected candidates. Median survival remains unreached; RMST at 60 months is 40.57 months.

1 Identity Enrichment

Layer 2 starts from the Layer-1 source population and applies buyer-identity evidence in priority order: exact SIRET reconciliation, same validated SIREN, historical name-and-department alias reconciliation, then Layer-1 name fallback. No fuzzy matching is used. Conflicts between BOAMP-native SIRET-derived SIREN and external enrichment are flagged rather than overwritten.

Table 1: Current source-level buyer identity coverage.

Identity source	Source rows
NAME_FALLBACK	1,233
UNIQUE_NAME_DEPARTMENT_ALIAS	919
DIRECT_BOAMP_SIRET	826
DIRECT_ENRICHMENT_NOTICE_JOIN	354
DIRECT_BOAMP_SIRET_CONFLICT_FLAGGED	48
Sources with known SIREN in Layer 2	2,147

2 Layer Comparison

3 Interpretation

Layer 2 increases blocking coverage from 59.3% to 64.1% and the balanced event rate from 29.7% to 35.1%. This is a material identity-sensitivity result. It should not be read as evidence that Layer 2 is more correct in the legal sense, because no real BOAMP renewal ground truth exists. Added links have weaker text evidence than links already shared with Layer 1: median s_{text} is 0.087 for enrichment-added links versus 0.209 for shared links.

Table 2: Layer-1 versus Layer-2 current outputs.

Quantity	Layer 1	Layer 2
Eligible source contracts	3,380	3,380
Unique buyer keys	819	808
Sources with at least one candidate	2,005	2,165
Zero-candidate sources	1,375	1,215
Candidate pairs	10,862	13,524
Balanced accepted links	1,003	1,188
Balanced event rate	29.7%	35.1%
POTENTIAL-tier links	327	430
Median link text similarity	0.207	0.191
Median link margin	0.113	0.092

Table 3: Source-level transition from Layer 1 to Layer 2.

Transition status	Source count
Unlinked in both layers	2,188
Same selected candidate	962
Added by enrichment	189
Changed candidate	37
Removed by enrichment	4

Table 4: Survival handoff comparison.

Quantity	Layer 1	Layer 2
Rows	3,380	3,380
Events	1,003	1,188
Censoring rate	70.3%	64.9%
Survival at 24 months	0.6908	0.6332
Median survival	Not reached	Not reached
RMST at 60 months	43.51	40.57
Cox hazard ratio for log duration	0.3173	0.3258
Log-normal AFT concordance	0.6231	0.6203

The enriched layer therefore answers a bounded question: how much do survival and linkage conclusions move when validated buyer identity is made less fragmented? It does not provide a verified recall correction.

4 Current Source of Truth

- `data/processed/enriched/enriched_sources.csv`;
- `data/processed/enriched/enriched_candidate_pairs.csv`;
- `data/processed/enriched/enriched_links_balanced.csv`;
- `data/processed/comparison/layer_link_comparison.csv`;
- `data/processed/comparison/layer_comparison_summary.csv`;
- `reports/tables/survival_layer_conclusions_comparison.csv`.